Genuine Multipartite Entanglement Analysis

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Model                              Ground-state properties          GME summary
-----                              ----------------------           -----------
Hubbard ring (1-D, PBC)            E0                : -2.60008132  Extracted n-spin RDM : n = 5
                                   E1                : -2.60008132
System parameters                                                   Purity:
-----------------                  <S2>              : 0.750000       gamma = Tr(rho^2)
Sites (L)          : 5             S                 : 0.5000
Electrons (Ne)     : 5             <Sz>              : 0.250000     Separable cut criterion:
Hopping (t)        : 1.0           Multiplicity 2S+1 : 2.0            gamma_A = 1  <->  rho = rho_A x rho_B
Interaction (U)    : 4.0           Spin state        : Doublet (S=1/2)
Mode               : uniform                                       C_GME (Troiani et al.):
n-body order       : 5             Double occupancy                   C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))
                                   ----------------
                                   <D> total         : 0.459102     Convex roof (mixed):
                                   <D> / site        : 0.091820       C_GME(ρ) = min Σ pᵢ C_GME(|ψᵢ))

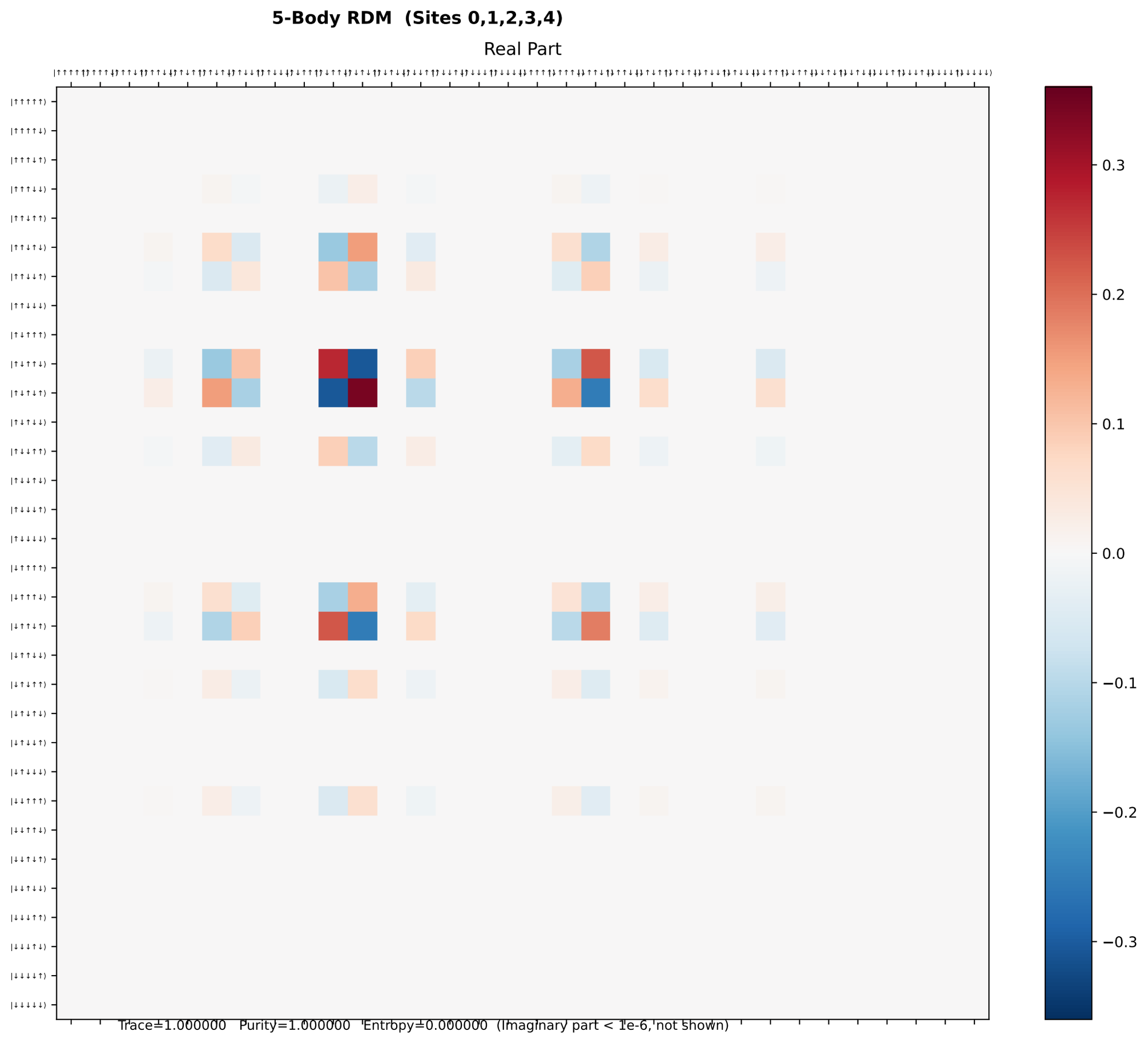

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```

Summary: All 5-Body RDM Clusters

System: L=5, Ne=5, U=4.0, t=1.0 | E_0=-2.600081

Sites	Purity	C_GME	C_LB	C_cayl	C_expo	C_UB	Status
(0, 1, 2, 3, 4)	1.000000	0.496333	0.000000	0.496333	0.496333	0.496333	GME
(1, 2, 3, 4, 0)	1.000000	0.496333	0.000000	0.496333	0.496333	0.496333	GME
(2, 3, 4, 0, 1)	1.000000	0.496333	0.000000	0.496333	0.496333	0.496333	GME
(3, 4, 0, 1, 2)	1.000000	0.496333	0.000000	0.496333	0.496333	0.496333	GME
(4, 0, 1, 2, 3)	1.000000	0.496333	0.000000	0.496333	0.496333	0.496333	GME

Legend:
  Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_cayl = unitary optimization - Cayley Transform |C_expo= unitary optimization - Expoenetial Map | BEST
  C_UB = Upper bound (eigendecomp) | Status: GME =has GME | NO GME=separable | MIXED=mixed state


5-Body RDM (Sites 0,1,2,3,4)
Real Part
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 0,1,2,3,4)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -1.681571e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  -------------|--------------|------------
  (0,)         | (1, 2, 3, 4) | 0.62109914
  (0, 1)       | (2, 3, 4)    | 0.77570535
  (0, 2)       | (1, 3, 4)    | 0.48594088
  (0, 3)       | (1, 2, 4)    | 0.32113729
  (0, 4)       | (1, 2, 3)    | 0.32832759
  (0, 1, 2)    | (3, 4)       | 0.87682669
  (0, 1, 3)    | (2, 4)       | 0.42085618
  (0, 1, 4)    | (2, 3)       | 0.43357804
  (0, 2, 3)    | (1, 4)       | 0.28510065
  (0, 2, 4)    | (1, 3)       | 0.27956910
  (0, 3, 4)    | (1, 2)       | 0.57073601
  (0, 1, 2, 3) | (4,)         | 0.52552231
  (0, 1, 2, 4) | (3,)         | 0.50899047
  (0, 1, 3, 4) | (2,)         | 0.75535749
  (0, 2, 3, 4) | (1,)         | 0.54458614

STATUS: GME EXISTS
------------------------------------

C_GME = 0.49633318
GME confirmed across ALL cuts.
lower bound [eqs.6-9]
  ----------------------------------

  C_exact = 0.49633318
  C_LB    = 0.00000000
  Gap     = 0.49633318
```

```
Convex Roof Analysis
====================================

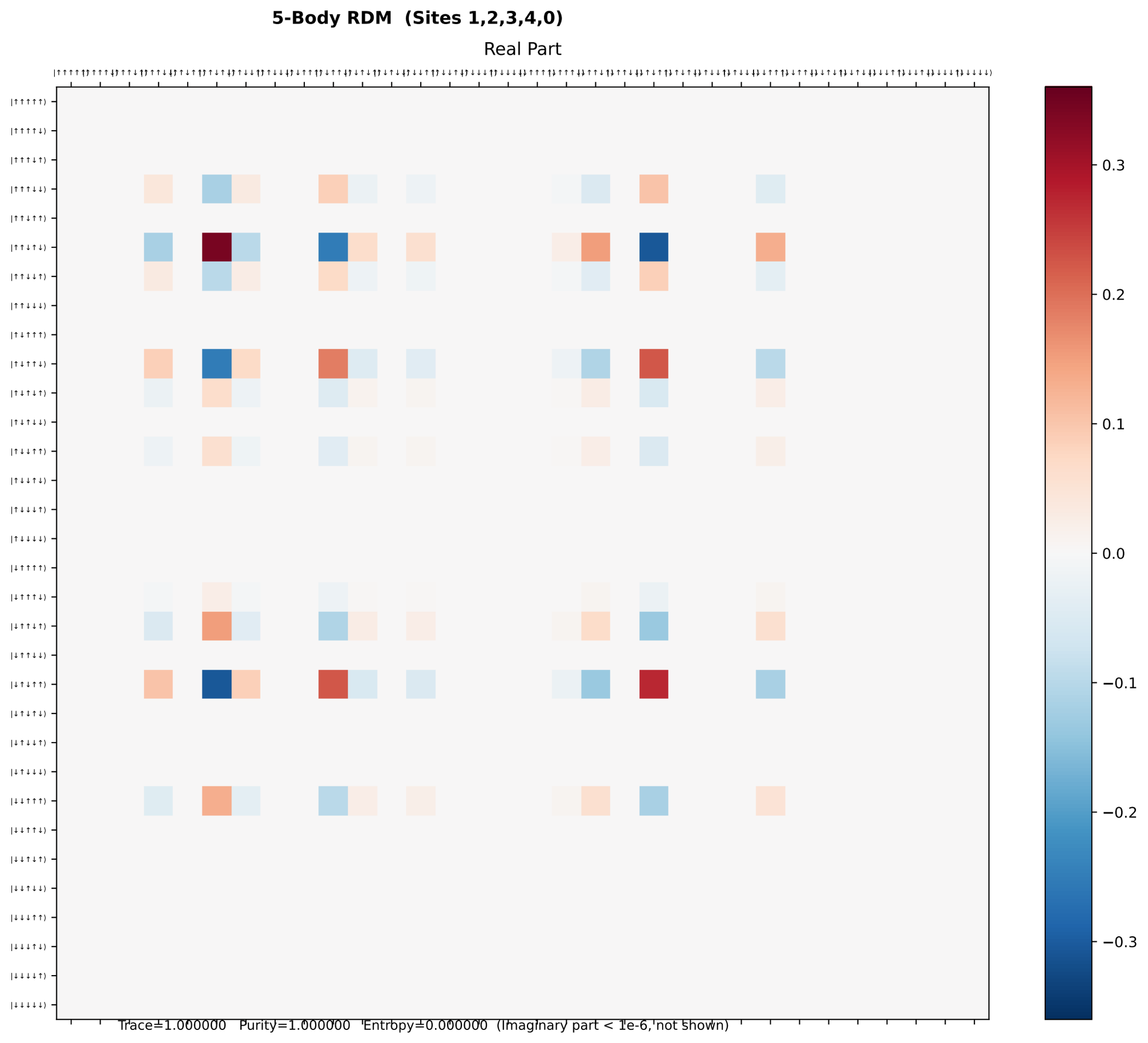
  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.49633318   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.49633318   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.49633318

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```


5-Body RDM (Sites 1,2,3,4,0)
Real Part
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 1,2,3,4,0)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -3.613134e-17
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  -------------|--------------|------------
  (1,)         | (2, 3, 4, 0) | 0.52552231
  (1, 2)       | (3, 4, 0)    | 0.32832759
  (1, 3)       | (2, 4, 0)    | 0.28510065
  (1, 4)       | (2, 3, 0)    | 0.42085618
  (1, 0)       | (2, 3, 4)    | 0.87682669
  (1, 2, 3)    | (4, 0)       | 0.43357804
  (1, 2, 4)    | (3, 0)       | 0.27956910
  (1, 2, 0)    | (3, 4)       | 0.57073601
  (1, 3, 4)    | (2, 0)       | 0.32113729
  (1, 3, 0)    | (2, 4)       | 0.48594088
  (1, 4, 0)    | (2, 3)       | 0.77570535
  (1, 2, 3, 4) | (0,)         | 0.50899047
  (1, 2, 3, 0) | (4,)         | 0.75535749
  (1, 2, 4, 0) | (3,)         | 0.54458614
  (1, 3, 4, 0) | (2,)         | 0.62109914

STATUS: GME EXISTS
------------------------------------

C_GME = 0.49633318
GME confirmed across ALL cuts.
lower bound [eqs.6-9]
  ----------------------------------

  C_exact = 0.49633318
  C_LB    = 0.00000000
  Gap     = 0.49633318
```

```
Convex Roof Analysis
====================================

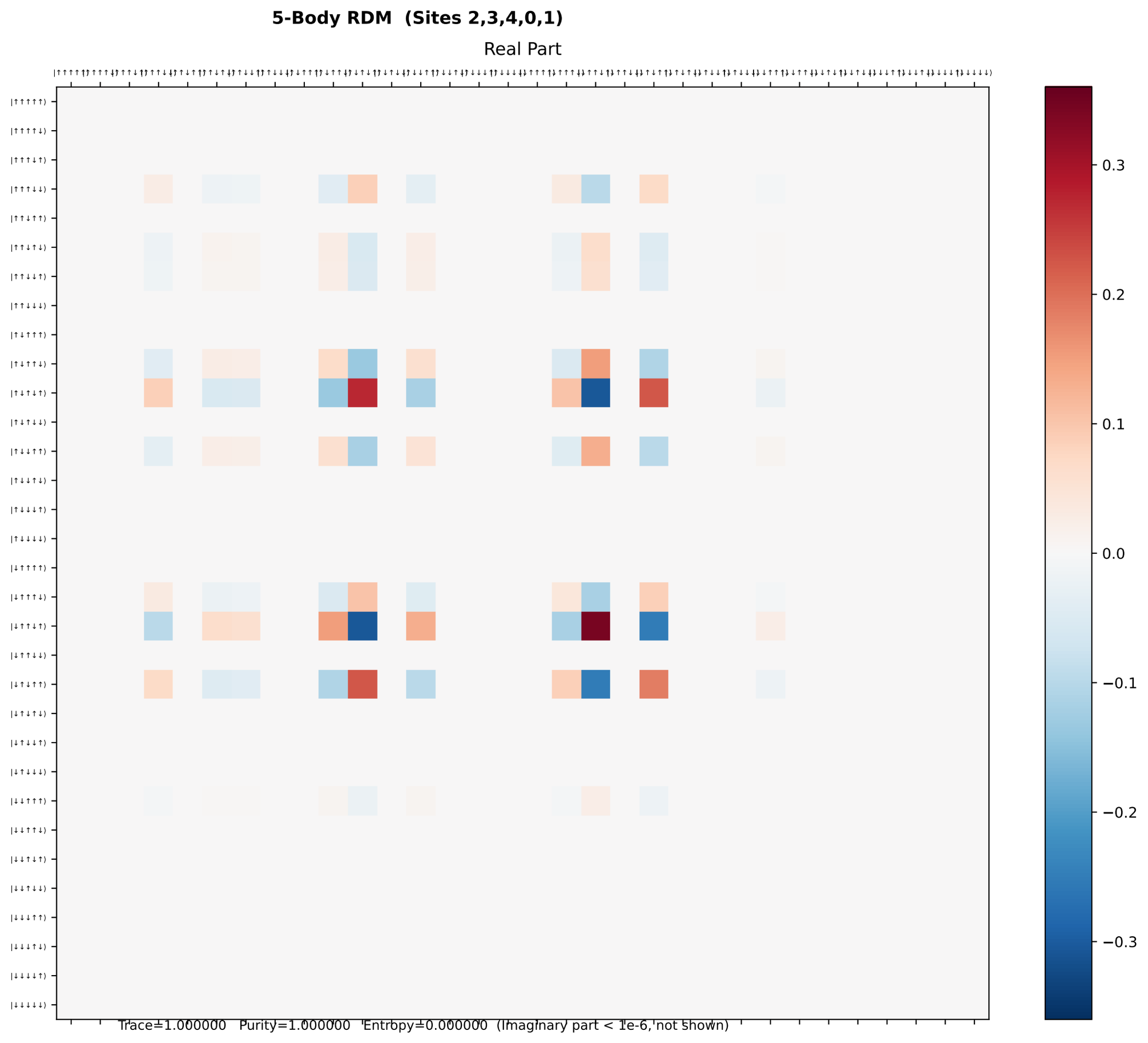
  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.49633318   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.49633318   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.49633318

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```


5-Body RDM (Sites 2,3,4,0,1)
Real Part
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 2,3,4,0,1)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -3.590841e-17
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (2,)         | (3, 4, 0, 1) | 0.50899047
  (2, 3)       | (4, 0, 1)    | 0.87682669
  (2, 4)       | (3, 0, 1)    | 0.32113729
  (2, 0)       | (3, 4, 1)    | 0.27956910
  (2, 1)       | (3, 4, 0)    | 0.43357804
  (2, 3, 4)    | (0, 1)       | 0.57073601
  (2, 3, 0)    | (4, 1)       | 0.48594088
  (2, 3, 1)    | (4, 0)       | 0.77570535
  (2, 4, 0)    | (3, 1)       | 0.42085618
  (2, 4, 1)    | (3, 0)       | 0.28510065
  (2, 0, 1)    | (3, 4)       | 0.32832759
  (2, 3, 4, 0) | (1,)         | 0.75535749
  (2, 3, 4, 1) | (0,)         | 0.54458614
  (2, 3, 0, 1) | (4,)         | 0.62109914
  (2, 4, 0, 1) | (3,)         | 0.52552231

STATUS: GME EXISTS
------------------------------------

C_GME = 0.49633318
GME confirmed across ALL cuts.
lower bound [eqs.6-9]
  ----------------------------------

  C_exact = 0.49633318
  C_LB    = 0.00000000
  Gap     = 0.49633318
```

```
Convex Roof Analysis
====================================

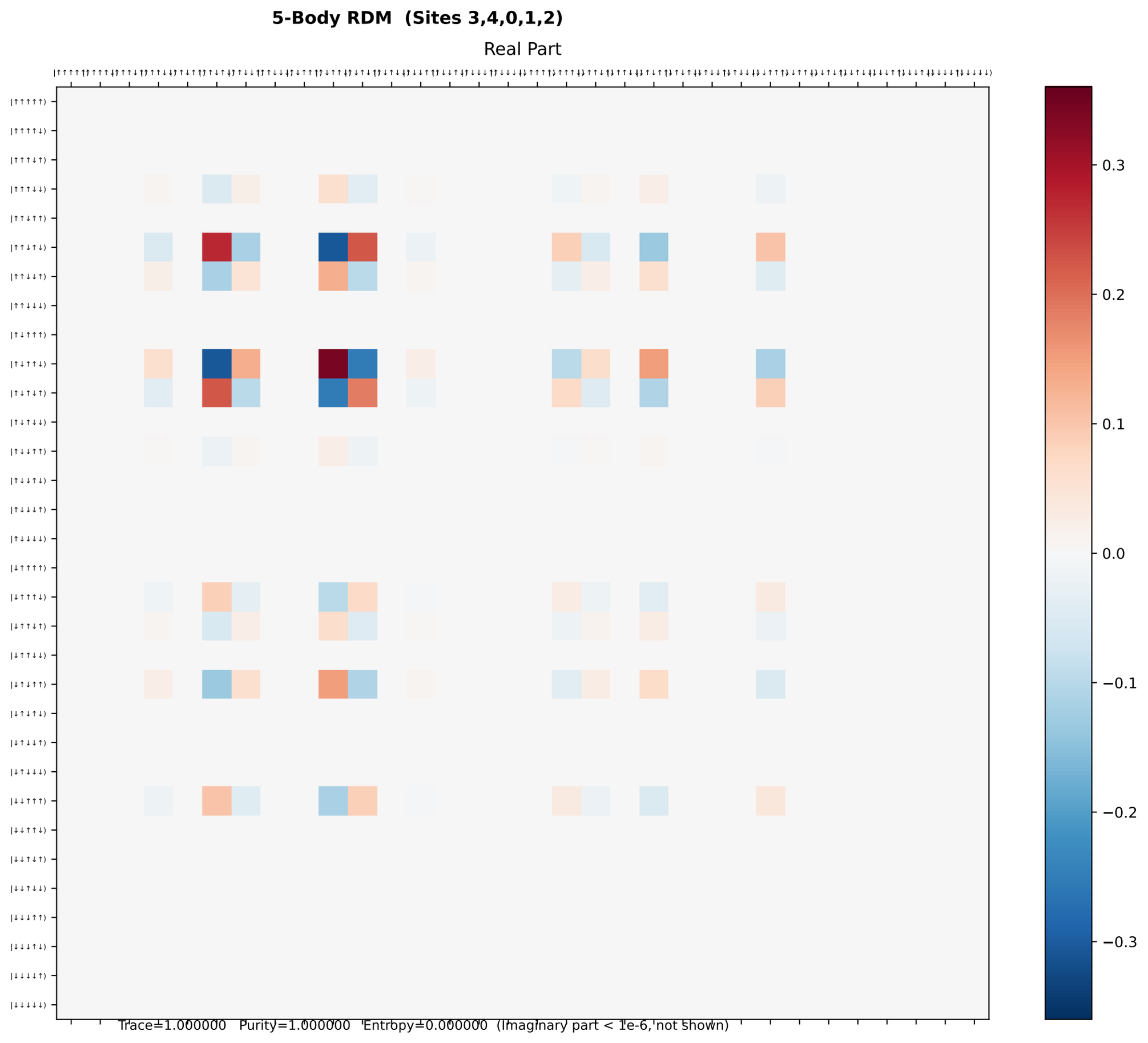
  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.49633318   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.49633318   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.49633318

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```


5-Body RDM (Sites 3,4,0,1,2)
Real Part
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 3,4,0,1,2)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -8.670812e-17
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  -------------|--------------|------------
  (3,)         | (4, 0, 1, 2) | 0.75535749
  (3, 4)       | (0, 1, 2)    | 0.43357804
  (3, 0)       | (4, 1, 2)    | 0.42085618
  (3, 1)       | (4, 0, 2)    | 0.48594088
  (3, 2)       | (4, 0, 1)    | 0.57073601
  (3, 4, 0)    | (1, 2)       | 0.77570535
  (3, 4, 1)    | (0, 2)       | 0.28510065
  (3, 4, 2)    | (0, 1)       | 0.32832759
  (3, 0, 1)    | (4, 2)       | 0.27956910
  (3, 0, 2)    | (4, 1)       | 0.32113729
  (3, 1, 2)    | (4, 0)       | 0.87682669
  (3, 4, 0, 1) | (2,)         | 0.54458614
  (3, 4, 0, 2) | (1,)         | 0.62109914
  (3, 4, 1, 2) | (0,)         | 0.52552231
  (3, 0, 1, 2) | (4,)         | 0.50899047

STATUS: GME EXISTS
------------------------------------

C_GME = 0.49633318
GME confirmed across ALL cuts.
lower bound [eqs.6-9]
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  C_exact = 0.49633318
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  Gap     = 0.49633318
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Convex Roof Analysis
====================================

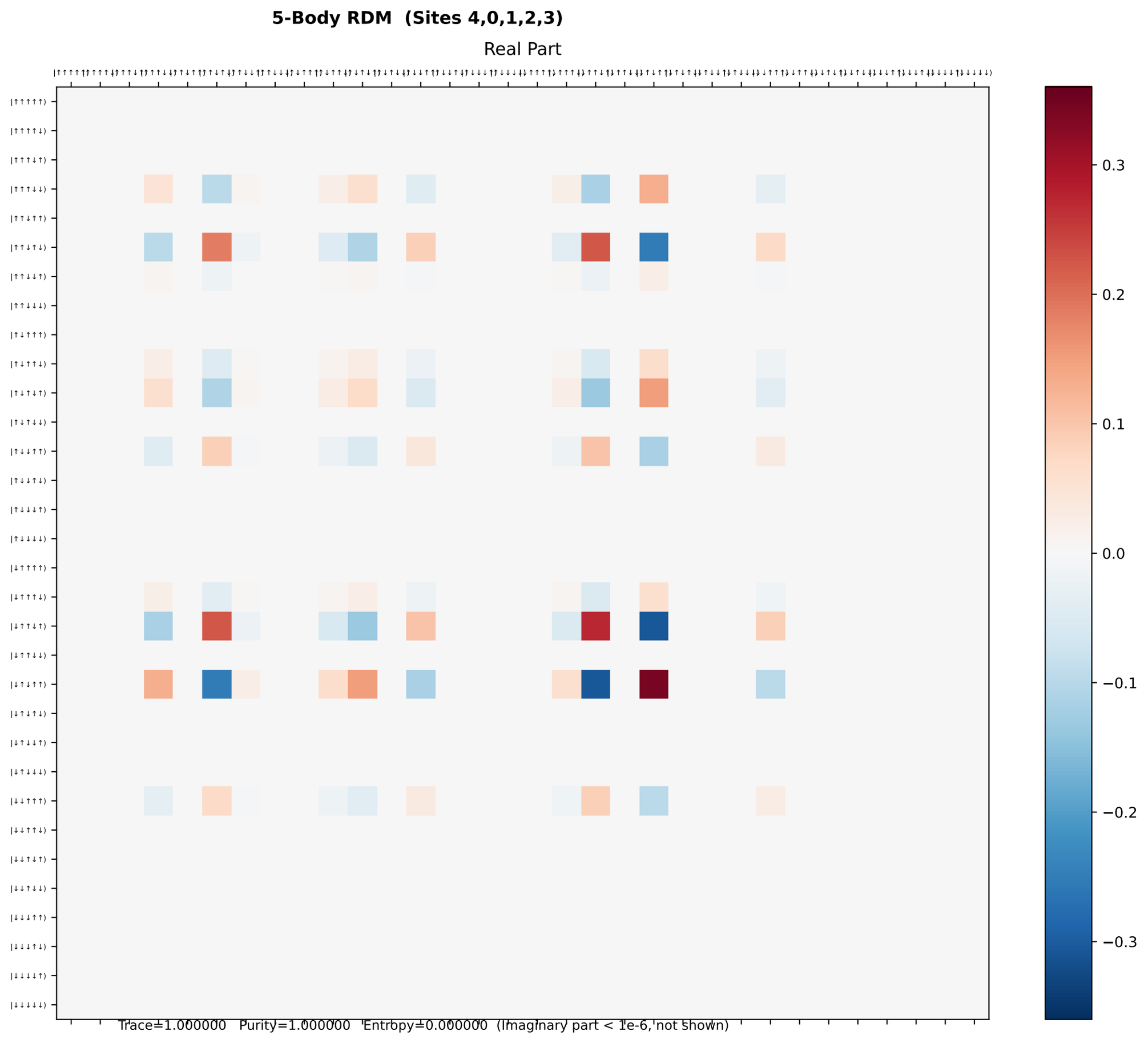
  Lower Bounds:
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    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.49633318   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.49633318   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.49633318

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```


5-Body RDM (Sites 4,0,1,2,3)
Real Part
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 5-Body RDM (Sites 4,0,1,2,3)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -1.249933e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  -------------|--------------|------------
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  (4, 0)       | (1, 2, 3)    | 0.57073601
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  (4, 2)       | (0, 1, 3)    | 0.28510065
  (4, 3)       | (0, 1, 2)    | 0.77570535
  (4, 0, 1)    | (2, 3)       | 0.32832759
  (4, 0, 2)    | (1, 3)       | 0.32113729
  (4, 0, 3)    | (1, 2)       | 0.87682669
  (4, 1, 2)    | (0, 3)       | 0.48594088
  (4, 1, 3)    | (0, 2)       | 0.42085618
  (4, 2, 3)    | (0, 1)       | 0.43357804
  (4, 0, 1, 2) | (3,)         | 0.62109914
  (4, 0, 1, 3) | (2,)         | 0.52552231
  (4, 0, 2, 3) | (1,)         | 0.50899047
  (4, 1, 2, 3) | (0,)         | 0.75535749

STATUS: GME EXISTS
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C_GME = 0.49633318
GME confirmed across ALL cuts.
lower bound [eqs.6-9]
  ----------------------------------

  C_exact = 0.49633318
  C_LB    = 0.00000000
  Gap     = 0.49633318
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.49633318   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.49633318   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.49633318

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```